



Hookworms — *Ancylostoma duodenale* & *Necator americanus*

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which larval stage is infective, and how does it enter the body?
2. Why is hookworm the commonest cause of iron-deficiency anaemia in endemic countries?
3. What is the drug of choice and how is it dosed?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Two hookworms — discriminator grid

	<i>A. duodenale</i>	<i>N. americanus</i>
Name	Old World hookworm	New World hookworm
Distribution	More endemic in parts of China, India, North Africa, sub-Saharan Africa, a few areas of the Americas	Predominant worldwide ; highest in sub-Saharan Africa, tropical Americas, China, SE Asia
Buccal capsule	Large, oval ; 2 pairs of claw-like teeth (ventral) + 1 pair of knob-like teeth (dorsal)	Small, round ; 2 pairs of cutting plates (ventral) + 1 pair (dorsal)
Blood loss / worm / day	0.15–0.26 mL (heavy infection: 10–20 mL/day)	0.03 mL
Severity	More serious	Less serious

Shared: adults in the small intestine (commonly jejunum), attached by the buccal cavity; live 2–3 years; female lays 10,000–25,000 eggs/day; **man is the only reservoir**; ~500 million infected. *A. ceylanicum* is mainly in cats (human cases in Malaysia and Asia).

Stage	Feature
Adult	Cylindrical, greyish-white, curved with the head bent (hook shape); male 8 mm with a copulatory bursa (13 rays, 3 lobes); female 12 mm
Egg	60 × 40 µm; thin transparent shell; 7–8 blastomeres ; clear space between ovum and shell
Rhabditiform larva	First stage; hatches from egg; feeding , motile; 200–300 µm; bulbed esophagus
Filariform larva	Infective stage ; non-feeding; 500–600 µm; shorter esophagus

Buccal capsules: *N. americanus* (plates) vs *A. duodenale* (teeth).

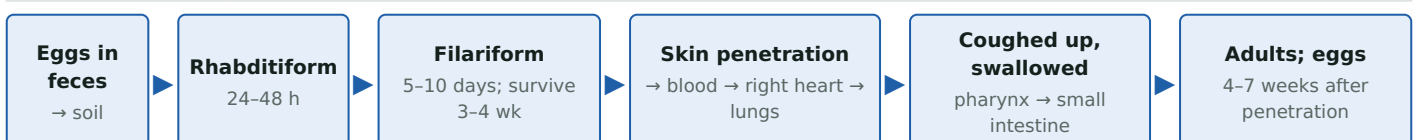
Egg with 7–8 blastomeres.



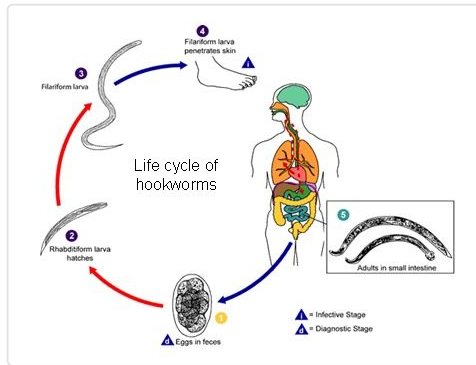
Adult worms.

2 How infection happens and the life cycle

Route	Notes
Cutaneous	Filariform larvae penetrate skin (feet, hands, buttocks) — most common
Oral	Ingestion of larvae with contaminated soil — less frequent
Trans-mammary / trans-placental	Rare / very rare



Life cycle in a single host (man); soil-transmitted helminth.



Life cycle of hookworms (infective stage = filariform larva; diagnostic stage = eggs in feces).

3 Pathogenesis — why the anaemia

Cause	Consequence
Adult attaches by buccal teeth, cuts vessels, feeds on blood and tissue fluid	Blood loss (direct absorption + bleeding from sites); anticoagulant factors prolong clotting time
Heavy, prolonged infection	Iron-deficiency anaemia (can be fatal if iron intake is poor); protein loss → hypoproteinaemia and oedema
Serine protease inhibitor	Interferes with pancreatic-enzyme digestion → malabsorption, growth retardation
Pregnancy	Iron deficiency + malnutrition harm intrauterine growth, birth weight and maternal survival
Larvae	Ground itch (allergic) at the entry site; lung passage → minute haemorrhage, leukocyte infiltration, eosinophilia , leukocytosis

4 Clinical picture by stage

Stage	Features
Skin	Ground itch — earliest; between the toes; intense itching, oedema, erythema, rash; secondary bacterial infection
Lung	Low-grade fever, mild cough, dyspnoea, pharyngitis, occasionally haemoptysis; pneumonia / bronchitis if severe
Gut	Low-grade fever, nausea, vomiting, diarrhoea, flatulence, abdominal discomfort; iron-deficiency anaemia + hypoalbuminaemia are the major manifestations

20–40% of cases in endemic areas are asymptomatic.

5 Immunity and immune evasion

Molecule	Action
Neutrophil inhibitory factor	Cysteine-rich glycoproteins: block adhesion of activated neutrophils to endothelium; inhibit H ₂ O ₂ release
Proteases	Elastin-cleaving metalloproteases cleave eotaxin → less eosinophil recruitment; <i>N. americanus</i> secretions cleave IgA
C-type lectins	Compete with host lectins for ligands (P-selectin, CD23)
Antioxidants	Superoxide dismutase and glutathione-S-transferase neutralise leukocyte oxidative burst

Repeated exposure does not necessarily give protection; antibodies do not correlate with resistance — so infection intensity often **rises with age** in endemic areas.

6 Diagnosis · treatment · prevention · creeping eruption

	Detail
Diagnosis	Suggested by eosinophilia + hypochromic microcytic anaemia ; eggs in stool (concentration for light infection; Kato-Katz counts eggs to estimate intensity); stool culture for larvae (Harada-Mori)

Treatment	Mebendazole = drug of choice (adults and larvae), 100 mg twice daily for 3 days; alternatives: pyrantel pamoate, thiabendazole, levamisole
Prevention	Sanitary faeces disposal, treat the infected, latrines, avoid walking barefoot , health education
Creeping eruption	<i>A. caninum</i> / <i>A. braziliense</i> (dog and cat hookworms): larvae cannot become adults in man and wander in the superficial skin (feet, legs, thigh, buttock); may reach the lung

P Must-remember pearls

- **Filariform larva** penetrating skin (feet) is the infective stage; cycle: skin → blood → lungs → swallowed → small intestine.
- Adults **attach and suck blood** → **iron-deficiency (hypochromic microcytic) anaemia** and hypoalbuminaemia.
- ***A. duodenale***: teeth, 0.15–0.26 mL/day, more serious. ***N. americanus***: cutting plates, 0.03 mL/day, worldwide predominant.
- **Ground itch** (skin) → cough / eosinophilia (lung) → anaemia (gut). Diagnose by **eggs in stool** (Kato-Katz to quantify).
- **Mebendazole 100 mg twice daily × 3 days**; prevent by latrines and wearing shoes.

Q Test yourself

1. A barefoot farmer has itchy, red swelling between his toes; weeks later he is anaemic with eosinophilia. Which route best explains the larval journey?

- A. Skin → blood → right heart → lungs → pharynx → swallowed → small intestine
 B. Mouth → stomach → liver → bile duct
 C. Skin → lymphatics → brain
 D. Anus → colon → appendix

2. Which finding separates *A. duodenale* from *N. americanus*?

- A. Blood loss of 0.15–0.26 mL per worm per day vs 0.03 mL
 B. Eggs with 7–8 blastomeres vs none
 C. Infective stage is the rhabditiform larva
 D. Adults live in the colon

3. Which test estimates the intensity of hookworm infection?

- A. Perianal cellophane tape
 B. Kato-Katz egg count
 C. Serology for IgE
 D. Chest X-ray

4. What is the drug of choice for hookworm, and how is it dosed?

- A. Praziquantel single dose
 B. Mebendazole 100 mg twice daily for 3 days
 C. Niclosamide 2 g over 2 days
 D. Mebendazole 100 mg once, repeated at 2 weeks

5. Itchy serpiginous tracks appear on the foot after contact with a dog or cat hookworm larva. Why does no adult worm develop?

- A. Man is an unsuitable host; larvae wander in superficial skin
 B. Larvae cannot penetrate skin
 C. The larvae are killed in the lung
 D. The adult stage lives only in the appendix

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Filariform larvae penetrate skin, travel via blood to the lungs, are swallowed and mature in the small intestine.
2 — A — *A. duodenale* causes more blood loss per worm and more serious disease; both share eggs and larval stages.
3 — B — Kato-Katz counts eggs in a known amount of stool.
4 — B — Mebendazole 100 mg twice daily for 3 days; pinworm uses a single dose repeated at 2 weeks.
5 — A — *A. caninum* and *A. braziliense* cause creeping eruption; man is not a suitable host.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam